

BÁO CÁO BÀI TẬP VỀ NHÀ NGÀY 09/06/2026

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1. Mã nguồn ftp_client

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#include <arpa/inet.h>
#include <netdb.h>
#include <stdbool.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/socket.h>
#include <sys/types.h>
#include <unistd.h>

int main() {
    char *server_host = "lebavui.io.vn";
    int control_port = 21;
    char buffer[2048];

    // phân giải tên miền thành IP
    struct hostent *he = gethostbyname(server_host);
    if (he == NULL) {
        perror("gethostbyname() failed");
        return 1;
    }

    char *server_ip = inet_ntoa(*(struct in_addr *)he->h_addr_list[0]);
    printf("IP của server: %s\n", server_ip);

    // kết nối kênh control port 21
    int control_sock = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
    struct sockaddr_in server_addr = {0};
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server_addr.sin_family = AF_INET;
server_addr.sin_port = htons(control_port);
server_addr.sin_addr.s_addr = inet_addr(server_ip);

if (connect(control_sock, (struct sockaddr *)&server_addr,
            sizeof(server_addr)) < 0) {
    perror("connect() port 21 failed");
    return 1;
}

// doc xau chao cua FTP server
int n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// gui username
send(control_sock, "USER user_20225241\r\n",
      strlen("USER user_20225241\r\n"), 0);
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// gui password
send(control_sock, "PASS 524111\r\n", strlen("PASS 524111\r\n"), 0);
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// chuyen sang che do bi dong (PASV) de mo kenh du lieu
send(control_sock, "PASV\r\n", strlen("PASV\r\n"), 0);
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';

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printf("%s", buffer);

// phan tich thong tin ip va port tu phan hoi cua lenh PASV
int h1, h2, h3, h4, p1, p2;
char *pasv_ptr = strchr(buffer, '(');
sscanf(pasv_ptr, "(%d,%d,%d,%d,%d,%d)", &h1, &h2, &h3, &h4, &p1, &p2);
int data_port = p1 * 256 + p2; // tinh data port

// mo socket data va ket noi den data_port
int data_sock = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in data_addr = {0};
data_addr.sin_family = AF_INET;
data_addr.sin_port = htons(data_port);
data_addr.sin_addr.s_addr = inet_addr(server_ip);
connect(data_sock, (struct sockaddr *)&data_addr, sizeof(data_addr));

// gui lenh LIST qua socket control
send(control_sock, "LIST\r\n", strlen("LIST\r\n"), 0);

// doc phan hoi server
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// nhan du lieu danh sach file tu socket data
char list_buf[4096] = {0};
int total_list_bytes = 0;
while ((n = recv(data_sock, list_buf + total_list_bytes,
                sizeof(list_buf) - total_list_bytes - 1, 0)) > 0) {
    total_list_bytes += n;
}
list_buf[total_list_bytes] = '\0';

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close(data_sock); // dong socket data sau khi nhan xong danh sach

// doc phan hoi server
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// tim file question
char *q_ptr = strstr(list_buf, "question_");
if (q_ptr == NULL) {
    printf("Khong tim thay file question_XXXXXX.txt tren server.\n");
    close(control_sock);
    return 1;
}
char question_file[128];
sscanf(
    q_ptr, "%s",
    question_file); // sscanf tu dong cat chuoai tai khoang trang/xuong dong
printf("File question can tai ve: '%s'\n", question_file);

// gui lai lenh PASV de lay kenh data moi cho viec tai file
send(control_sock, "PASV\r\n", strlen("PASV\r\n"), 0);
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';

pasv_ptr = strchr(buffer, '(');
sscanf(pasv_ptr, "(%d,%d,%d,%d,%d,%d)", &h1, &h2, &h3, &h4, &p1, &p2);
data_port = p1 * 256 + p2;

// ket noi data port
data_sock = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
data_addr.sin_port = htons(data_port);

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connect(data_sock, (struct sockaddr *)&data_addr, sizeof(data_addr));

// gui lenh RETR de lay noi dung file question
char retr_cmd[256];
snprintf(retr_cmd, sizeof(retr_cmd), "RETR %s\r\n", question_file);
send(control_sock, retr_cmd, strlen(retr_cmd), 0);

// doc phan hoi server
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// nhan noi dung file
char file_content[1024] = {0};
int total_content_bytes = 0;
while ((n = recv(data_sock, file_content + total_content_bytes,
                sizeof(file_content) - total_content_bytes - 1, 0)) > 0) {
    total_content_bytes += n;
}
file_content[total_content_bytes] = '\0';
close(data_sock);

// doc phan hoi server
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// loai bo ky tu '\r' hoac '\n' trong noi dung
file_content[strcspn(file_content, "\r\n")] = 0;
printf("Noi dung file question: %s\n", file_content);

// tao ten file answer_xxxxxx.txt ung voi question_xxxxxx.txt

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char answer_file[128];
char *rand_part =
    strchr(question_file, '_');
snprintf(answer_file, sizeof(answer_file), "answer%s", rand_part);
printf("File answer se duoc tao ra: '%s'\n", answer_file);

// dao nguoc xau ky tu
int content_len = strlen(file_content);
char reversed_content[1024];
for (int idx = 0; idx < content_len; idx++) {
    reversed_content[idx] = file_content[content_len - 1 - idx];
}
reversed_content[content_len] = '\0';

// luu file answer
FILE *f = fopen(answer_file, "w");
fprintf(f, "%s", reversed_content);
fclose(f);
printf("Da luu file cuc bo voi noi dung dao nguoc: %s\n", reversed_content);

// goi PASV de lay kenh truyen du lieu cho viec upload
send(control_sock, "PASV\r\n", strlen("PASV\r\n"), 0);
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';

pasv_ptr = strchr(buffer, '(');
sscanf(pasv_ptr, "(%d,%d,%d,%d,%d,%d)", &h1, &h2, &h3, &h4, &p1, &p2);
data_port = p1 * 256 + p2;

// ket noi data port
data_sock = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
data_addr.sin_port = htons(data_port);

```

```
connect(data_sock, (struct sockaddr *)&data_addr, sizeof(data_addr));

// gui lenh STOR qua control socket
char stor_cmd[256];
snprintf(stor_cmd, sizeof(stor_cmd), "STOR %s\r\n", answer_file);
send(control_sock, stor_cmd, strlen(stor_cmd), 0);

// doc phan hoi server
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// day du lieu tu bien vao socket data de thuc hien upload
send(data_sock, reversed_content, strlen(reversed_content), 0);
close(data_sock);

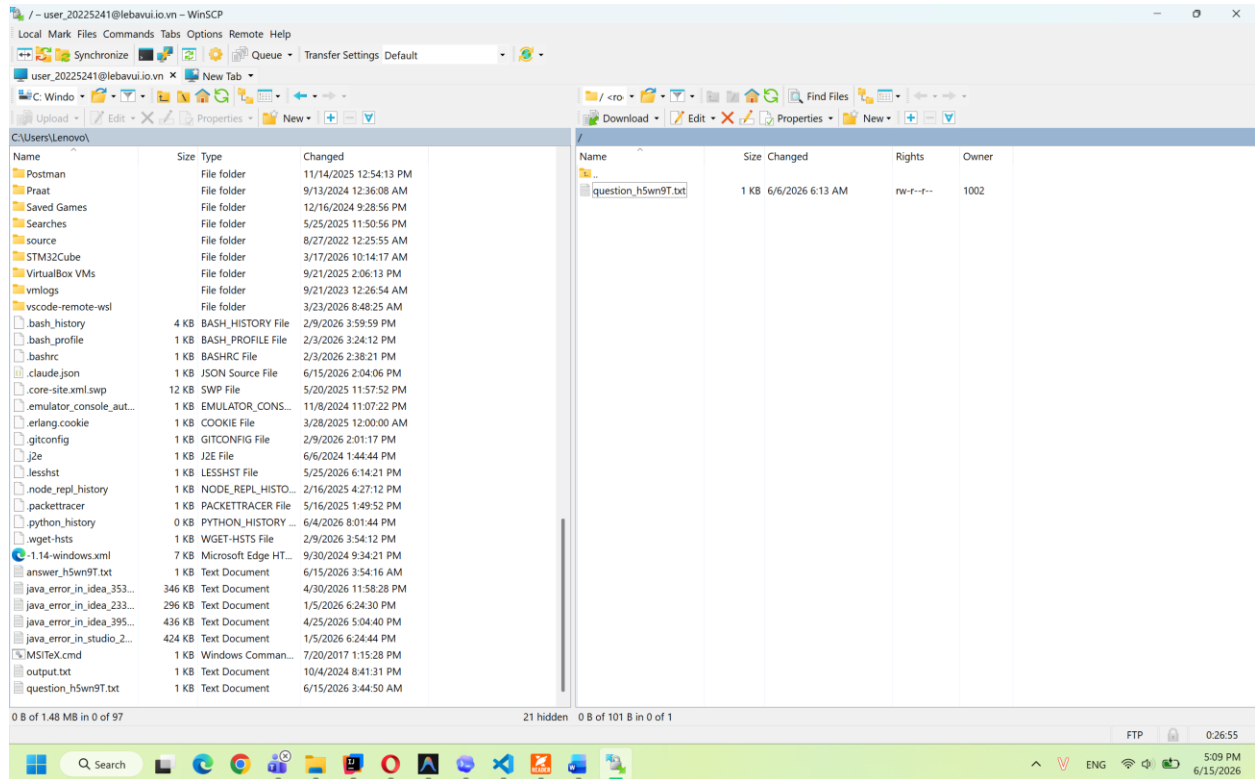
// doc phan hoi server
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

// dong ket noi
send(control_sock, "QUIT\r\n", strlen("QUIT\r\n"), 0);
n = recv(control_sock, buffer, sizeof(buffer) - 1, 0);
buffer[n] = '\0';
printf("%s", buffer);

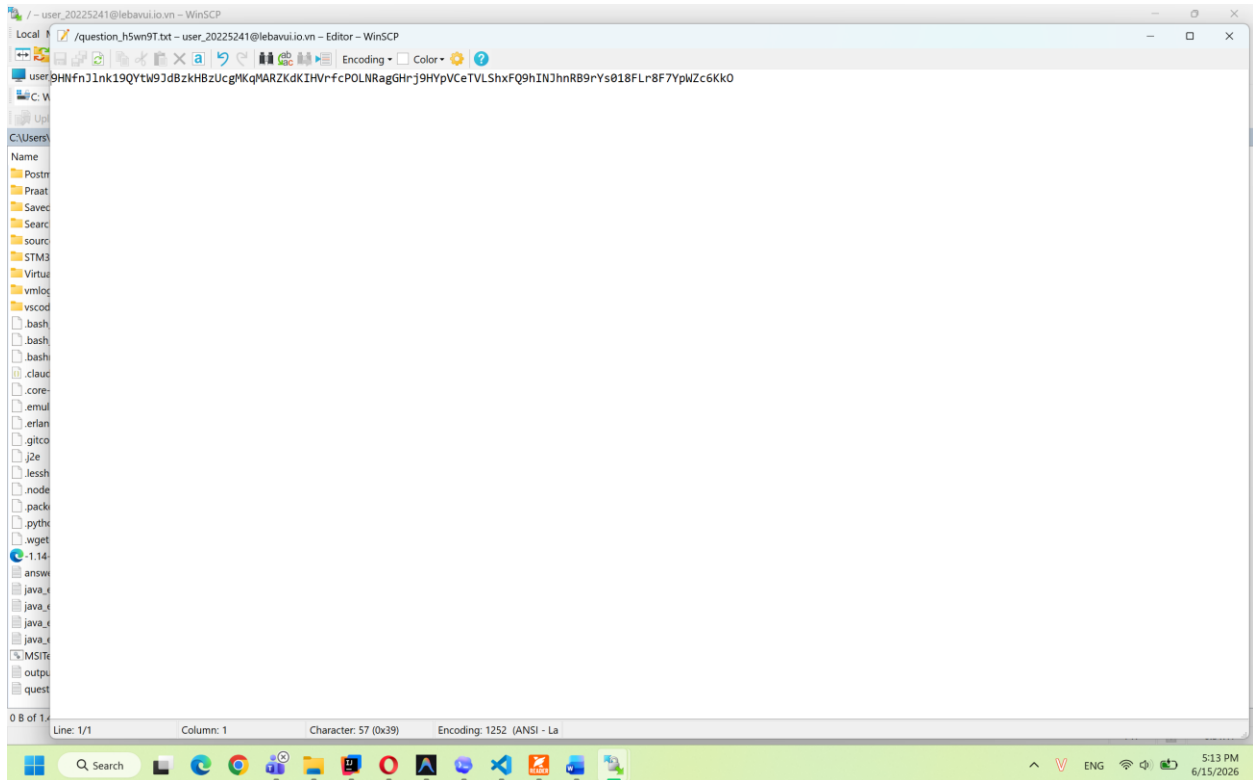
close(control_sock);
return 0;
}
```

2. Kết quả

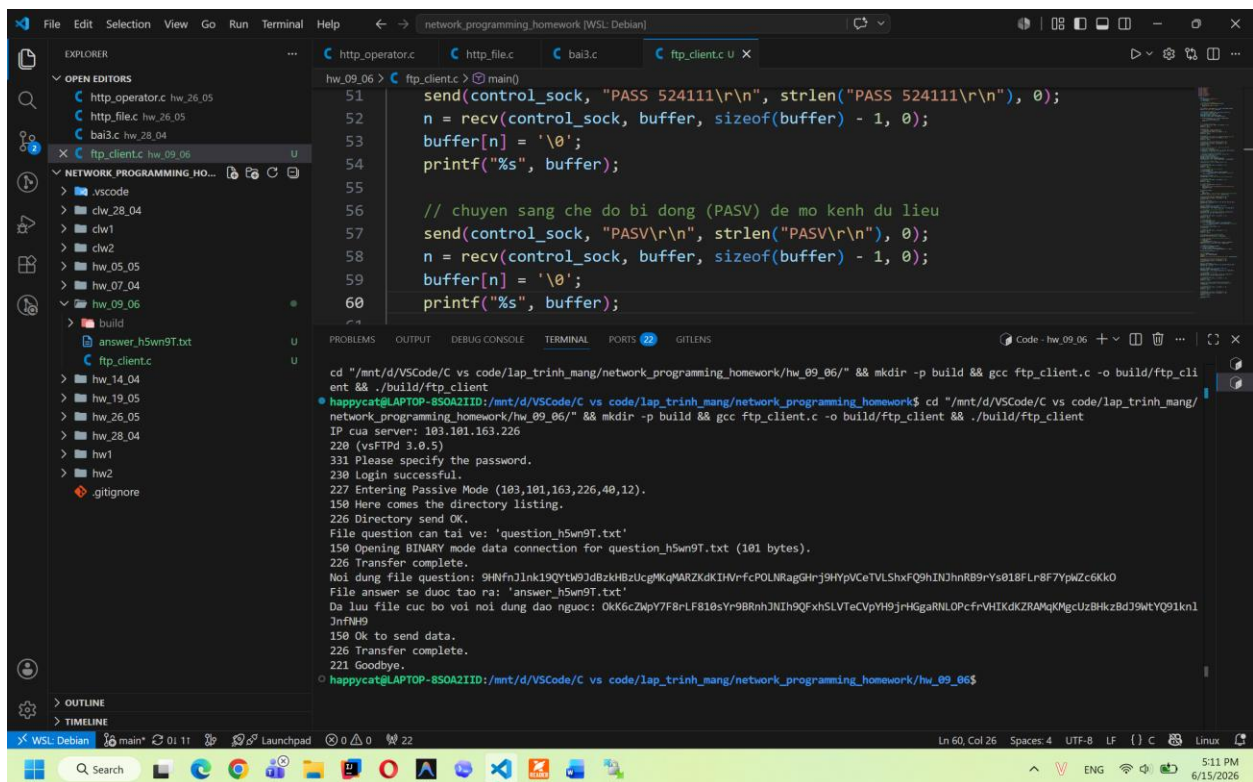
- Kết nối server lebavui.io.vn bằng WinSCP để kiểm tra kết quả
- Khi chưa chạy chương trình, trên server chỉ có 1 file



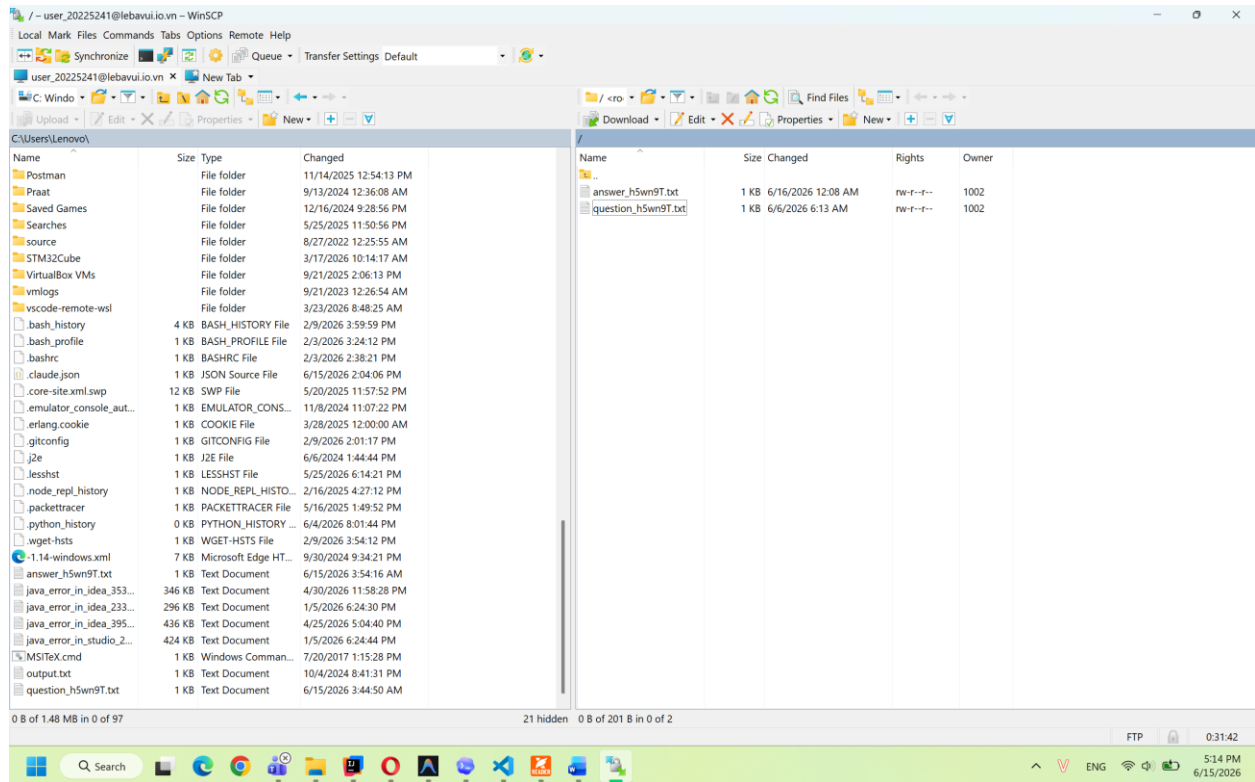
- Nội dung file question



- Chạy FTP client để đọc file question_h5wn9T.txt, tạo file answer_h5wn9T.txt với nội dung đảo ngược và upload lên server



- Kết quả trên server



- Nội dung file answer

